

Amazon RDS & Migration

Managed relational databases, availability and migration

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01 Amazon RDS

Amazon Relational Database Service (Amazon RDS) is a managed relational database service. AWS takes responsibility for much of the underlying database infrastructure and routine maintenance, while you focus on configuring and using the database.

What does managed actually mean?

Imagine your application needs a MySQL database. You could provision a server, install MySQL, maintain the operating system, configure backups and plan for hardware failure yourself. RDS moves much of that work to AWS.

Self-managed MySQL	Managed with RDS
Provision the server yourself	AWS provisions it
Install MySQL	AWS handles installation
Maintain the underlying OS	AWS handles it
Set up backup mechanisms	RDS provides automated backups
Handle hardware failures	AWS manages the infrastructure
Configure your own failover setup	RDS can provide automated failover with Multi-AZ
Patch everything yourself	RDS manages much of the patching process

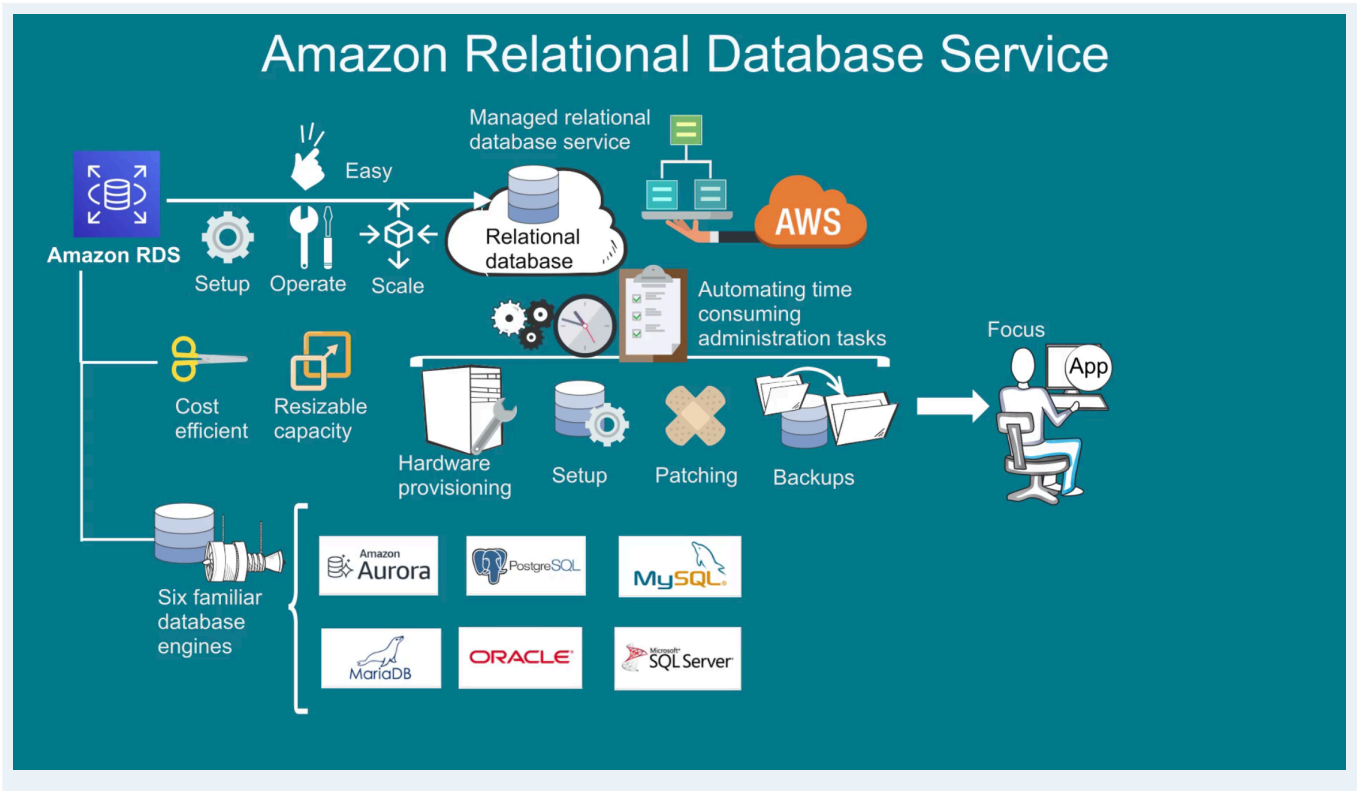
You still manage the database itself: tables, data, queries, users, permissions, indexes and the configuration choices exposed by the service.

Mental model

AWS operates and maintains much of the database platform; you focus on the data and how your application uses it.

Supported relational database engines

Database engine	What it is
Amazon Aurora	AWS-built relational database compatible with MySQL and PostgreSQL
MySQL	Popular open-source relational database
PostgreSQL	Open-source relational database with strong features and extensibility
MariaDB	Open-source database originally derived from MySQL
Oracle Database	Commercial enterprise relational database from Oracle
Microsoft SQL Server	Microsoft's commercial relational database
IBM Db2	IBM enterprise relational database



Why RDS is useful

Benefit	What it means
Easy to administer	AWS handles much of the infrastructure provisioning, installation and maintenance.
Easy to scale	Compute, memory and storage can be increased as the database grows.
Read replicas	Copies of the database can handle read requests and reduce load on the primary.
Managed infrastructure	AWS operates much of the underlying platform so you can focus on the application and data.

Reliability & performance

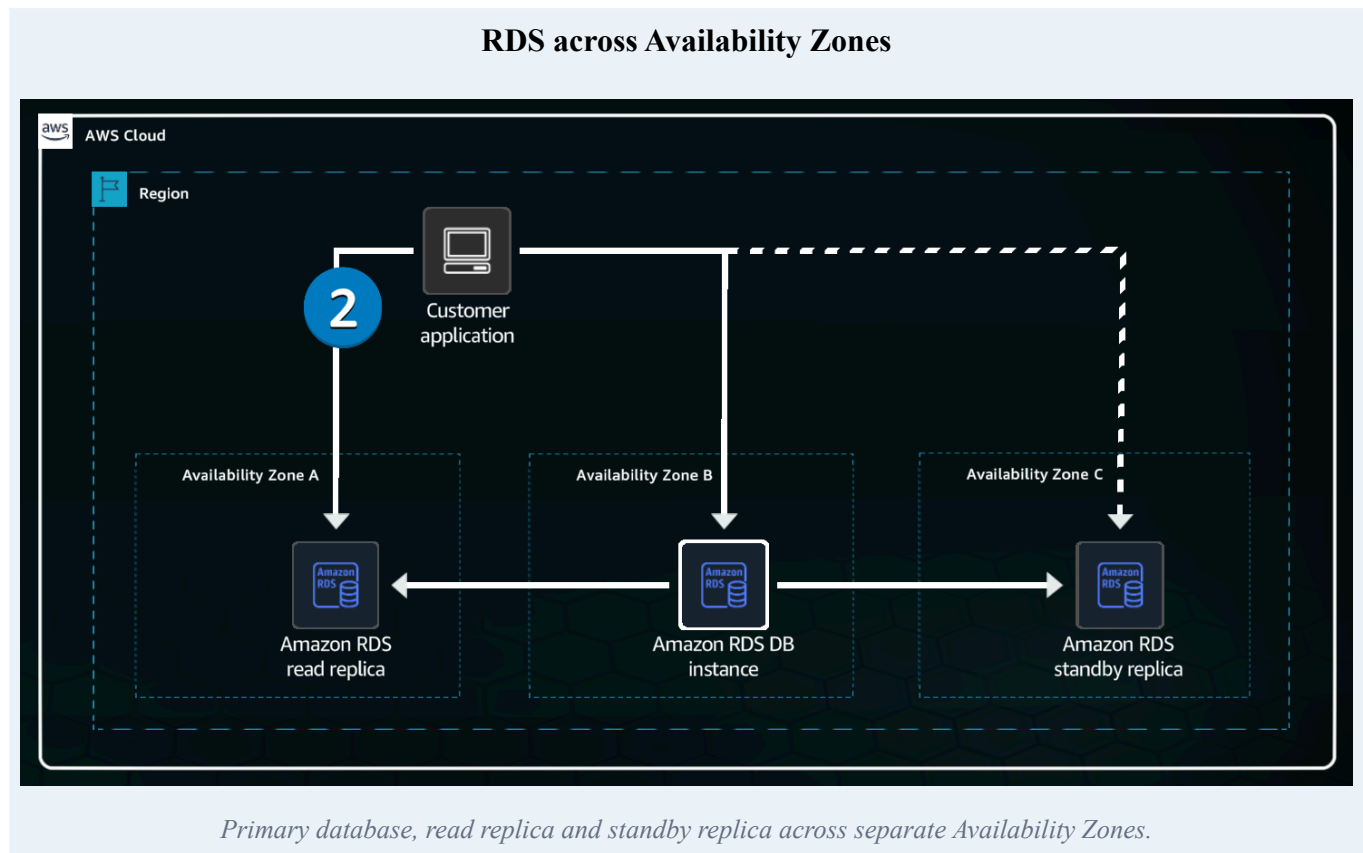
Benefit	What it means
Available & durable	RDS can replicate databases across Availability Zones.
Automated backups	RDS can automatically back up the database and support recovery.
Automatic host replacement	AWS can replace failed underlying infrastructure.
Quick provisioning	New RDS DB instances can be created without manually installing a database server.
Fast storage	SSD-backed storage options support demanding workloads.
Amazon Aurora	AWS relational engine designed for performance and availability with MySQL/PostgreSQL compatibility.

Summary

RDS automates infrastructure provisioning, backups, snapshots and host replacement to reduce database administration.

RDS across Availability Zones

Here the primary RDS database runs in Availability Zone B.
Additional database copies exist in other Availability Zones for different reasons.



Database	Location	Purpose
RDS DB instance	AZ B	Primary database used for normal reads and writes
Read replica	AZ A	Handles read traffic and reduces load on the primary
Standby replica	AZ C	Provides availability and can take over if the primary fails

The important difference

Read replica = performance. Standby replica = availability.

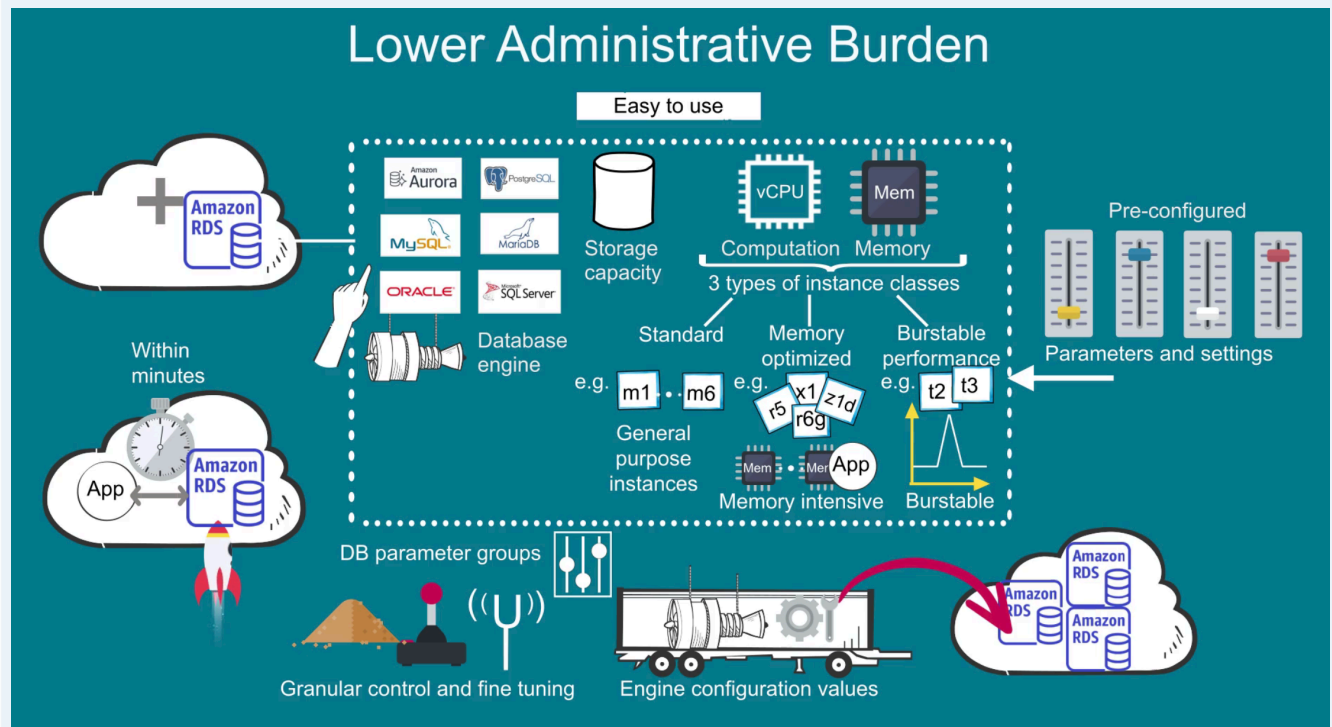
Lowering administrative burden

Wouldn't it be useful if AWS could handle much of the routine work for us: provisioning infrastructure, installing software, patching, backups, maintenance and recovery?

That's where Amazon RDS comes in.

RDS is a managed database service. AWS handles much of the underlying infrastructure and routine maintenance, while we still control how our database should be configured.

RDS instance configuration



So, what do we manage?

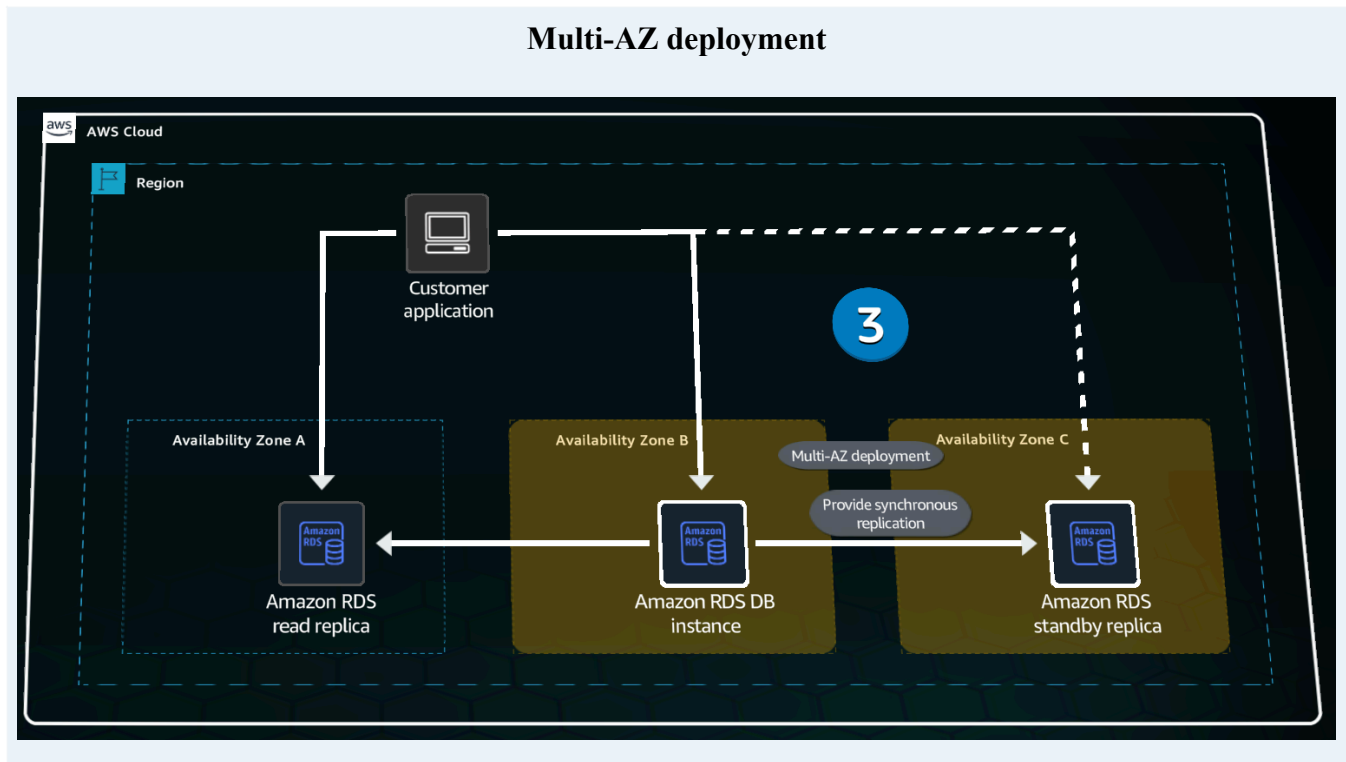
Using a managed service does not mean AWS makes every decision for us. Instead, the responsibility is divided.

We Decide	AWS manages
Which DB engine to use	Underlying DB infrastructure
How much compute and memory we need	Operating system and server maintenance
How much storage we need	Routine software patching
DB settings and configuration	Hardware failures and replacement
Backup and retention preferences	Performing automated backups

Availability & Durability

Multi-AZ deployment

High availability is achieved through Multi-AZ deployment, where RDS synchronously replicates data from the primary instance to a standby instance in a different Availability Zone.



Primary and standby form the high-availability deployment.

If the primary becomes unavailable, RDS can fail over to the standby. Writes normally go to the primary, and the DB instance endpoint remains the same so the application can recover without being reconfigured.

What does Multi-AZ mean?

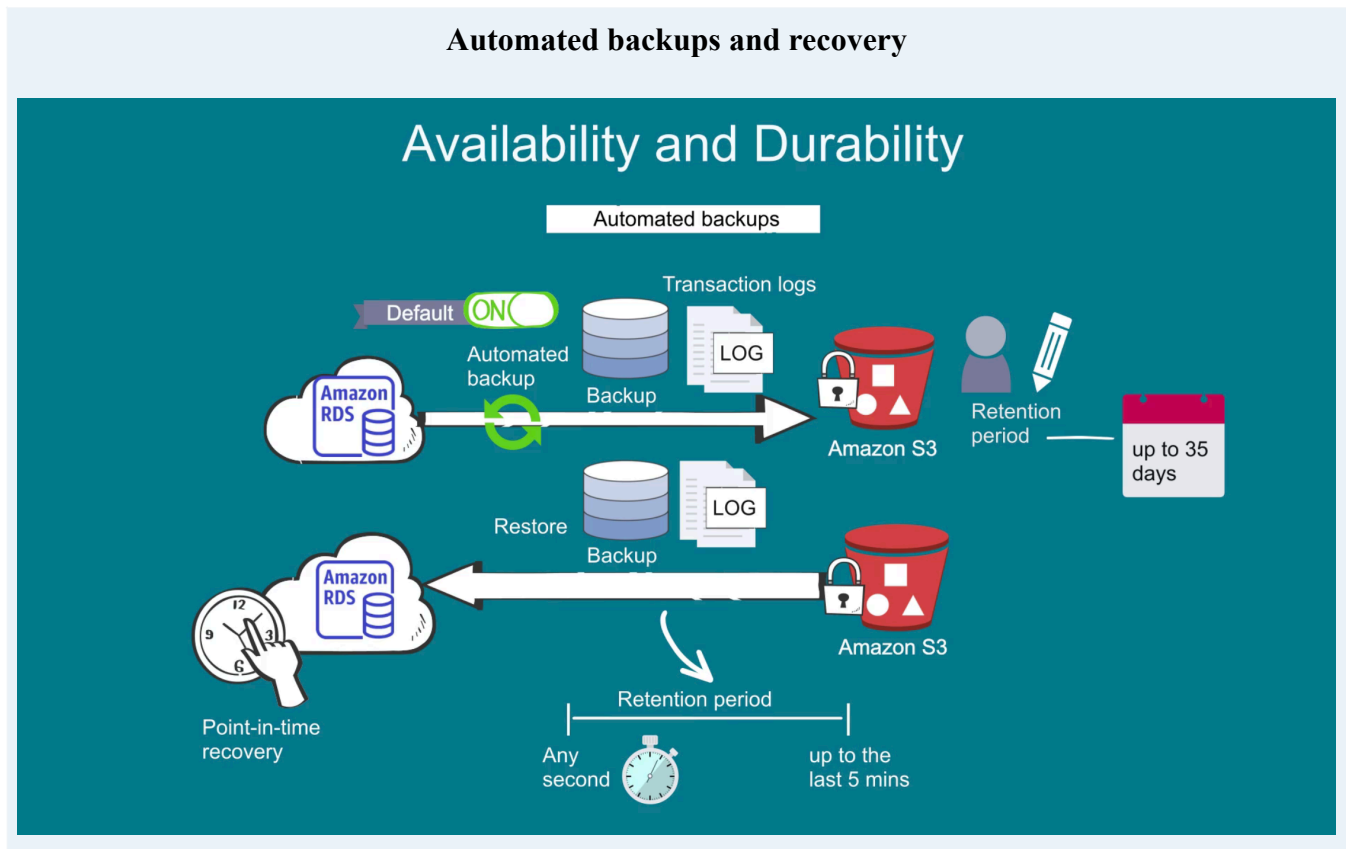
It sounds like it simply means using multiple Availability Zones, but in traditional RDS terminology Multi-AZ specifically refers to a high-availability deployment with a primary database and a standby in another AZ.

A read replica in another AZ helps with read performance, but it does not replace the standby used for automatic failover.

Automated backups

RDS can automatically back up the database and transaction logs for a user-defined retention period of up to 35 days. These backups support point-in-time recovery.

Point-in-time recovery lets you restore the DB instance to a specific moment within the retention window. Transaction logs are uploaded continuously, but the latest restorable time is typically a few minutes behind the current time.



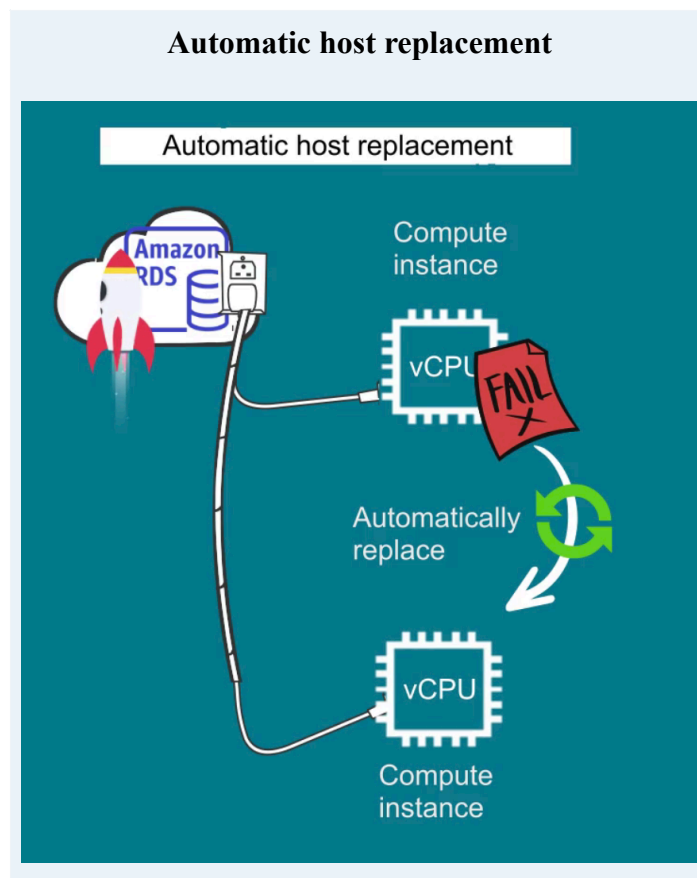
Database snapshots

A manual RDS snapshot saves the state of the database so you can return to it later. This is useful before major changes such as a schema migration.

- The first snapshot contains the data required to represent the database at that point.
- Later snapshots only need to store new or changed blocks that are not already stored.
- Billing is based on backup and snapshot storage actually consumed, not simply the full database size for every snapshot.
- Deleting an older snapshot may not immediately free all of its storage if newer snapshots still depend on shared blocks.

Automatic host replacement

An RDS database ultimately runs on AWS-managed compute infrastructure. If that underlying hardware fails, RDS can automatically replace the failed compute host instead of requiring you to provision a replacement yourself.



AWS manages replacement of failed underlying database infrastructure.

What to remember

- RDS manages much of the infrastructure and routine database administration.
- Multi-AZ is for high availability and automatic failover.
- Read replicas are primarily for scaling read traffic.
- Automated backups provide a recovery window; manual snapshots preserve chosen points in time.
- AWS manages underlying host failures while you remain responsible for your data and database design.

02 Practice Lab

Migrating a customer application database

Task: migrate a customer application database to Amazon RDS. The goal is to reduce routine database administration while preparing the database architecture for higher availability and better read performance.

In the practice lab we will

- Migrate the customer application database to an Amazon RDS DB instance.
- Use a single-AZ RDS instance as the starting deployment.
- Explore Multi-AZ deployment for high availability.
- Explore read replicas to improve read performance.

Practice goal

Move database infrastructure responsibility toward RDS, then use managed RDS features to improve availability and performance without managing the underlying database host yourself.

Before we start: EC2 & AMIs

Before creating our RDS database, let's quickly revisit Amazon EC2 and introduce Amazon Machine Images (AMIs).

EC2 recap

Amazon EC2 provides virtual servers in AWS. An EC2 instance gives you compute resources such as CPU and memory where applications and services can run. In this lab, an EC2 instance could run the web application that communicates with our RDS database.

What is an AMI?

An Amazon Machine Image (AMI) is a reusable template used to create EC2 instances. It can define the starting operating system, installed software, configuration and storage setup.

EC2 instance	AMI
A running virtual server	A template used to create a server
Runs applications	Defines the server's starting environment
Uses compute resources	Does not run by itself
Created from an AMI	Can create many EC2 instances

If several servers need the same starting environment, an AMI avoids configuring each one from scratch.

aws

Search

[Option+S]

United States (N. Virginia)

Account ID: 4086-8797-5267
AWSLabUser-fWaTcimghNCwrdshvcg8Aa/Sadf338...

EC2 > AMI Catalog

EC2

Dashboard

AWS Global View

Events

Instances

Images

AMI Catalog

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Settings

AMIs

Create Template with AMI

Launch Instance with AMI

sql

Quick Start AMIs (13)
Commonly used AMIs

My AMIs (0)
Created by me

AWS Marketplace AMIs (641)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

☐ Free tier only

☒ OS category

☐ All Linux/Unix

☐ All Windows

☒ Architecture

☐ 64-bit (Arm)

☐ 32-bit (x86)

sql (13 filtered, 13 unfiltered)

Canonical

Ubuntu

Verified provider

Ubuntu Server 22.04 LTS (HVM) with SQL Server 2022 Standard
ami-091b599f5f318ddd2 (64-bit (x86))
Microsoft SQL Server 2022 Standard edition on Ubuntu Server 22.04 LTS.
Platform: ubuntu Root device type: ebs Virtualization: hvm ENA enabled: Yes

Select

64-bit (x86)

Microsoft

Windows

Verified provider

Microsoft Windows Server 2025 with SQL Server 2025 Standard
ami-046d4b20db861eed2 (64-bit (x86))
Microsoft Windows 2025 Datacenter edition with SQL Server 2025 Standard.
Platform: windows Root device type: ebs Virtualization: hvm ENA enabled: Yes

Select

64-bit (x86)

AMIs provide reusable starting templates for EC2 instances.

Why this matters

A database can run directly on EC2, but the server and database environment remain largely your responsibility. RDS removes much of that administration.

Database on EC2	Amazon RDS
You manage the virtual server	AWS manages the underlying infrastructure
You install and configure the database	RDS provisions the database
You manage OS updates	AWS manages the underlying OS
You handle routine database patching	RDS can handle routine patching
You design and manage backups	RDS provides automated backups
You configure HA and failover	RDS can provide managed Multi-AZ failover
More control	Less administration

Setting up RDS

Select Aurora and RDS to begin creating a managed database. RDS can automate hardware provisioning, database setup, patching and backups.

The screenshot shows the AWS console search results for the term 'database'. The interface is dark-themed. At the top, there's a search bar with 'database Insights' and a button 'Ask Amazon Q'. Below the search bar, it says 'Search results for 'database''. The main section is titled 'Services results for 'database'' with a 'Back' link. There are five service cards listed: 1. 'Oracle Database@AWS' with a description 'Lift and shift Oracle Exadata workloads and rapidly innovate with the broad portfolio of AWS services'. 2. 'Database Migration Service' with a description 'Managed Database Migration Service'. 3. 'DynamoDB' with a description 'Managed NoSQL Database'. 4. 'AWS AppSync' with a description 'Real-Time Data Sync Using GraphQL for Mobile & Web Apps, Online or Offline'. 5. 'Aurora and RDS' with a description 'Managed Relational Database Service'. Each card has a star icon on the right. At the bottom, there's a 'Top features' section with links: 'Dashboard', 'Databases', 'Query Editor', 'Performance Insights', and 'Snapshots'.

Search results for 'database'

Services results for 'database'

[Back](#)

-  **Oracle Database@AWS**
Lift and shift Oracle Exadata workloads and rapidly innovate with the broad portfolio of AWS services
-  **Database Migration Service**
Managed Database Migration Service
-  **DynamoDB**
Managed NoSQL Database
-  **AWS AppSync**
Real-Time Data Sync Using GraphQL for Mobile & Web Apps, Online or Offline
-  **Aurora and RDS**
Managed Relational Database Service

Top features

[Dashboard](#) [Databases](#) [Query Editor](#) [Performance Insights](#) [Snapshots](#)

Open Aurora and RDS.



[Aurora and RDS](#) > Databases

Aurora and RDS



Dashboard

Databases

Query editor

CloudWatch Database Insights 

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Navigate to Databases.

No resources

No resources to display

Create database ▲

 Express configuration

Full configuration

Restore from S3

Choose Create database, then Full configuration.

Choose the database engine

Select the relational database engine required by the application.

Create database [Info](#)

Engine options
Engine type [Info](#)

☐ Aurora (MySQL Compatible) 	☐ Aurora (PostgreSQL Compatible) 	☐ MySQL 	☒ PostgreSQL 
☐ MariaDB 	☐ Oracle 	☐ Microsoft SQL Server 	☐ IBM Db2 

Choose the engine type.

For the lab, choose the Dev/Test template.

Templates
Choose a sample template to meet your use case.

☐ Production Use defaults for high availability and fast, consistent performance.	☒ Dev/Test This instance is intended for development use outside of a production environment.	☐ Sandbox To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.
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Templates provide sensible starting settings for different use cases.

Engine versions

RDS supports major and minor engine versions. The exact naming varies by engine, but the practical difference is simple:

Version type	Example	Usually includes
Major version	MariaDB 11.8	Larger feature, behavior and compatibility changes
Minor version	MariaDB 11.8.8	Bug fixes, security fixes and smaller improvements

A major-version upgrade deserves more planning and testing because it can affect compatibility, configuration and SQL behavior. Minor upgrades are generally less disruptive because they remain within the same major release family. Minor upgrades can often be automated during maintenance windows; major upgrades are planned and tested more carefully. Some engines can use paid RDS Extended Support beyond standard support.

Settings

Engine version [Info](#)
View the engine versions that support the following database features.

▼ **Hide filters**

☒ **Show only versions that support the Amazon RDS Optimized Writes** [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MariaDB 11.8.8 ▼

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

my-database

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Choose a supported engine version.

Credentials & instance configuration

Credentials management

RDS can manage master credentials with AWS Secrets Manager, avoiding hardcoded passwords and supporting rotation. For this lab, we keep things simple and use self-managed credentials.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - *most secure***
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ **Auto generate password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

.....

Password strength **Very strong**
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

.....

Choose how the database master credentials are managed.

Instance configuration

RDS instance classes provide different combinations of CPU, memory, storage and networking capacity.

Instance class	Best suited for
Standard (M classes)	Balanced CPU and memory for general workloads
Memory optimized (R/X classes)	Memory-intensive production databases
Burstable (T classes)	Smaller or variable workloads that occasionally need more CPU

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2.085 Mbps Network: Up to 5 Gbps

Storage configuration

An RDS database needs persistent storage. In this lab we use General Purpose SSD (gp3), a common balance of cost and performance.

Setting	What it means
Storage type	Storage technology and performance characteristics; here we use gp3 SSD
Allocated storage	Database capacity; here we allocate 20 GiB
IOPS	Storage read/write operations per second
Throughput	Amount of data transferred between database compute and storage per second

Storage

Storage type

Info

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)

Performance scales independently from storage

Allocated storage

Info

20

GiB

Minimum: 20 GiB. Maximum: 65 536 GiB

Provisioned IOPS

Info

3000

IOPS

Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GiB.

Storage throughput

Info

125

MiBps

Baseline storage throughput of 125 MiBps is included for allocated storage less than 400 GiB.

To provision additional IOPS and throughput, increase the allocated storage to 400 GiB or greater.

Additional storage configuration

Configure persistent database storage.

Storage autoscaling

If the database grows beyond its allocated storage, RDS Storage Autoscaling can automatically increase capacity when available space becomes low.

Additional storage configuration

Storage autoscaling

Info

Provides dynamic scaling support for your database's storage based on your application's needs.

Enable storage autoscaling

Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Maximum storage threshold

Info

Charges will apply when your database autoscales to the specified threshold

1000

GiB

Allocated storage value must be 22 GiB to 65 536 GiB

Autoscaling reduces the need to manually increase storage before space runs out.

AWS Cloud • Part IV

Availability, security & monitoring

Availability & durability

For high availability, RDS can maintain a synchronous standby instance in another Availability Zone and automatically fail over if the primary becomes unavailable.

Availability & durability

Multi-AZ deployment [Info](#)

- ☐ Do not create a standby instance
- ☒ Create a standby instance (recommended for production usage)
Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.

Multi-AZ adds a standby database for availability and failover.

Security with VPC

RDS databases can run inside a VPC so network access can be controlled using the same networking concepts covered earlier, including subnets, routing and security groups.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-03bd77cbe959170b5)
6 Subnets, 6 Availability Zones



After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

- ☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Place the database inside the appropriate VPC and control who can reach it.

Monitoring

Database Insights can collect performance information such as database load, query performance, resource-heavy queries and database activity over time.

Monitoring [Info](#)
Choose the monitoring configuration for this database. The Standard mode of Database Insights offers seven days of retention for detailed database metrics at no additional charge. The Advanced mode offers fleet-level monitoring and premium features, including the long-term retention of database metrics. Database Insights pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

- Fleet-level monitoring
- Integration with CloudWatch Application Signals
- Detailed database and per-query metrics (extended retention at no additional charge)

☒ Database Insights - Standard

- Detailed database and per-query metrics (free for 7 days of retention)
- Option to retain detailed metrics up to 24 months for an additional charge

☒ Collect detailed database and per-query metrics

Retention period
7 days (free) ▼

AWS KMS key [Info](#)
(default) aws/rds ▼

Monitoring helps explain how the database behaves under load.

Do not confuse the two

Historical performance metric retention is for monitoring and analysis. Backup retention controls how far back the database can be restored.

Additional configuration

Specify the initial database name required by the application.

▼ Additional configuration
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options
Initial database name [Info](#)
my_database
If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)
default.mariadb11.8 ▼

Option group [Info](#)
default:mariadb-11-8 ▼

Set the initial database name.

RDS can encrypt data at rest using AES-256, protecting stored data, disk I/O and snapshots.

☒ **Enable encryption**
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console.

Enable storage encryption when required.

Automatic minor-version upgrades can keep routine engine updates current during maintenance windows.

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version [documentation](#).

Minor updates can be automated.

Review the estimated cost

Before creating the database, review the monthly cost estimate and confirm that the selected instance, storage and availability options match the intended workload.

Estimated monthly costs

DB instance	198.56 USD
Storage	4.60 USD
Total	203.16 USD

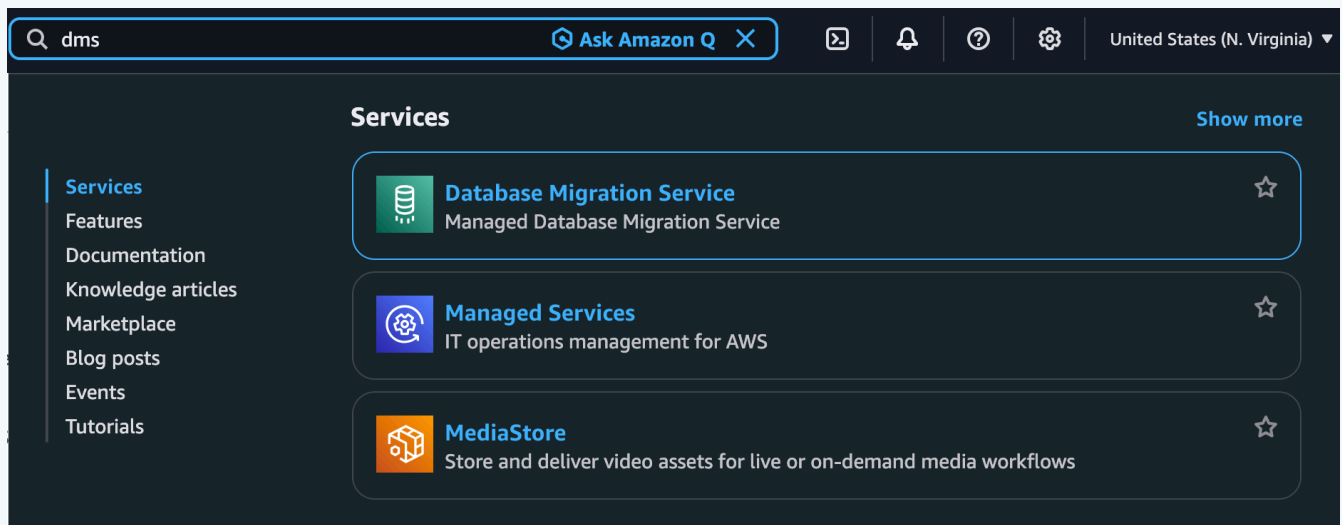
This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, IOs (if applicable), or data transfer.

Estimate your monthly costs for the DB Instance using the [AWS Simple Monthly Calculator](#).

Review the estimated monthly cost before deployment.

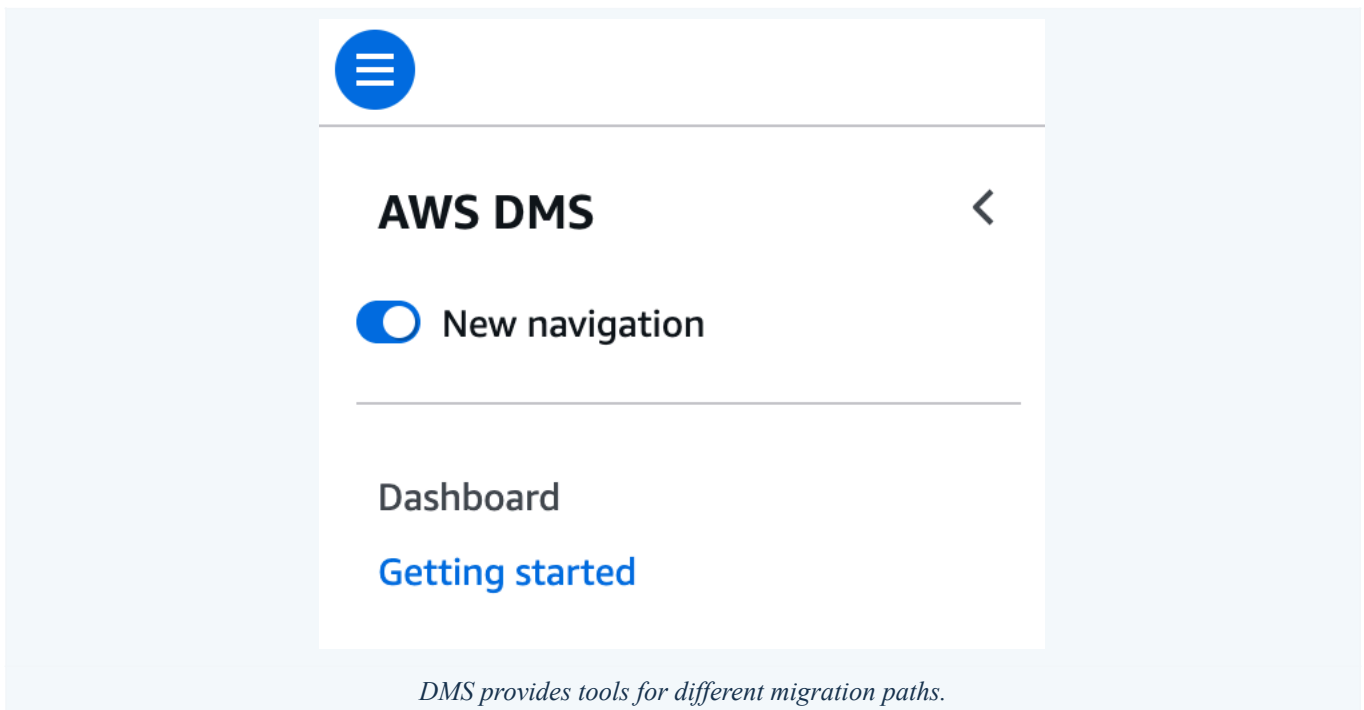
Database Migration Service

AWS Database Migration Service (AWS DMS) can move data between data stores while keeping the source database operational, helping minimise downtime during migration.



Open AWS Database Migration Service.

DMS supports migrations involving relational databases, data warehouses and NoSQL data stores, including migrations between different database engines.



DMS provides tools for different migration paths.

Choose a starting point based on where the data is today and where it needs to move.

Get started

To get started with DMS, choose a starting point to find out what tools are available for your data movement journey.

How you can use DMS

☒ Convert or move to managed



☐ Migrate or replicate



Convert your database schemas with integrated DMS Schema Convert tool or run performant like to like data migration with native database tooling. Choose **Go to landing page** to find more information.

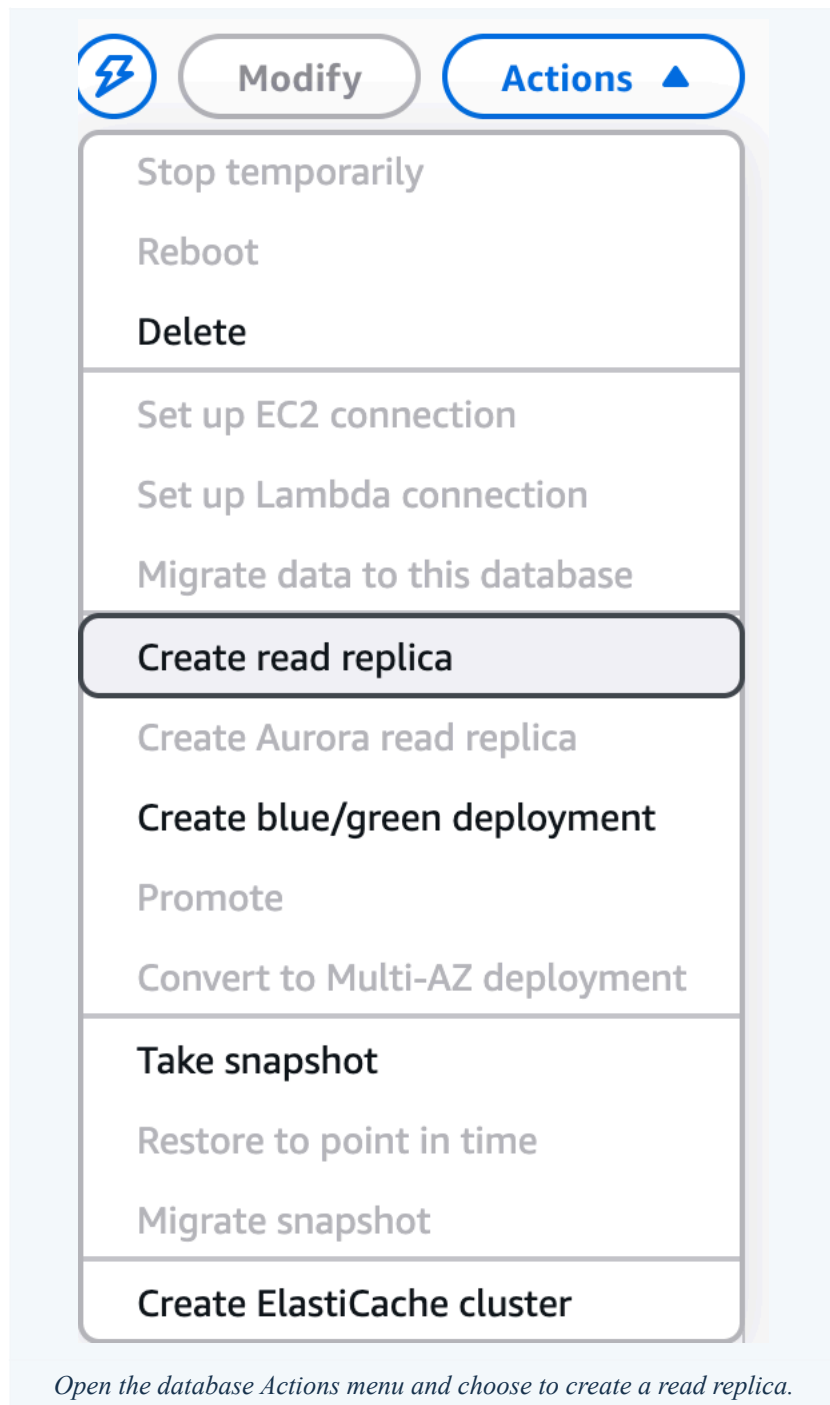
 You can find supported engine types, related documentation, or related tools.

[Go to landing page](#)

Select the appropriate data movement journey.

Create a read replica

A read replica creates another database instance that can serve read traffic, reducing load on the primary database. In this example, we create a read replica using a db.t3.xlarge instance.



Name the replica, review its configuration and create it.

Create read replica

You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.

Settings

Replica source

Source DB instance identifier

my-database
Role: Instance

DB instance identifier

This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).

my-database-read-replica

Name and create the read replica.

RDS then provisions the replica and begins synchronising it with the primary database.

Creating replica my-database-read-replica in US East (N. Virginia)

Your database might take a few minutes to launch.

1 0 0 0 1

Databases (2)

Group resources



Modify

Actions

Create database

Filter by databases

1

DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...
my-database	Modifying	Primary	MariaDB	SECOND	us-east-1e
my-database-read-replica	Creating	Replica	MariaDB	SECOND	-

Wait for the read replica to become available.

The lab's final mental model

EC2 can host the application, RDS manages the relational database, Multi-AZ protects availability, read replicas scale reads, and DMS helps move existing data into the managed environment.

What to Remember

Amazon RDS provides a managed way to run relational databases. We still design the database and choose the resources it needs, while AWS handles much of the underlying infrastructure, maintenance, patching and recovery work.

Different features solve different problems:

Need more compute or memory? Choose a different DB instance class.

Running out of storage? Increase storage or use storage autoscaling.

Too many read queries? Add a read replica.

Need protection from an Availability Zone failure? Use Multi-AZ.

Need to recover data? Use automated backups, point-in-time recovery or snapshots.

Need to understand database performance? Use monitoring such as Database Insights.

Need to move an existing database? AWS DMS can help migrate the data.

These choices also affect **cost**.

Larger DB instances and more storage cost more, while features such as Multi-AZ and read replicas add additional database resources that you pay for. Backup storage, monitoring and data transfer can also contribute to the final cost depending on how they are used.

The goal is therefore not to choose the largest or most resilient configuration possible, but to match the database resources, availability and performance to what the application actually needs.

RDS reduces the work of running a database, but the choices we make determine both its capabilities and its cost.